

Home Studio

Site workflow handbook

How the interior-design marketplace works end to end: what visitors, customers and administrators can do, how each flow moves through the application, and how it is set up, seeded, themed and tested.

Version	1.1
Date	28 September 2026
Stack	Laravel 12 on PHP 8.2+, Blade, Tailwind CSS 4, Alpine.js, Vite 6 (Node 20.14+), SQLite or MySQL
Local URL	http://localhost:8000
Admin	admin@home.studio / Admin123!
Demo customer	demo@example.com / Demo123!

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1. What the site does

Home Studio sells home interior design image collections. Every **design** belongs to one **design style** (Japandi, Zen Minimalist, Modern Hanok and so on) and one **room type** (living room, bedroom, kitchen and so on) and carries a gallery of images shot from different angles, each with its own title and description. A design can also carry **360° panoramas**, grouped by floor into a walk-through tour. A design is either **free** or has a one-time **price**.

Free designs are fully visible to everyone. For premium designs, everyone sees the cover image and the list of angles, but the remaining images and the 360° tour stay locked until the visitor signs in and pays once. After that the design is unlocked for that account forever, with full-resolution downloads.

The site ships with two complete visual themes (Calm and Neon) that an administrator can preview and switch, two interface languages (English and Simplified Chinese), and an admin panel that manages the whole catalogue, orders, customers, themes and site settings.

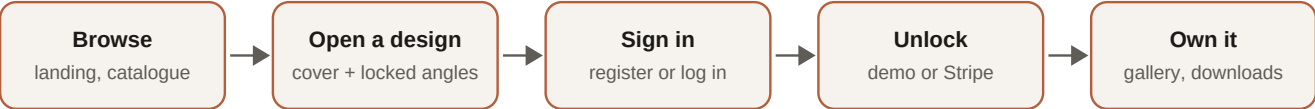


Figure 1. The core customer journey.

2. Roles and permissions

Role	Who	Can do
Visitor	Anyone, not signed in	Browse everything, search and filter, open free designs and download them, see cover + locked angles of premium designs, like and share, switch language and (Calm theme) appearance.
Customer	Registered account (role customer)	Everything a visitor can, plus unlock premium designs by paying, see the full gallery and download all images of unlocked designs, manage profile and see orders in My Studio.
Admin	Account with role admin	Everything a customer can, plus the admin panel: dashboard, designs and galleries, styles, room types, orders, customers, themes and settings. Admins see every gallery unlocked and can download any design.

Access rules are enforced on the server: the **auth** middleware guards customer routes, the **admin** middleware (EnsureAdmin) guards the whole /admin area, and the design page and download endpoint check *Design::unlockedFor(user)* before rendering any locked image URL. Locked image URLs are never sent to the browser, so they cannot be found by inspecting the page.

3. Architecture and request flow

The application is a conventional Laravel monolith: routes map to controllers, controllers load Eloquent models and render Blade views, and a small amount of Alpine.js adds interactivity (menus, search suggestions, carousel, lightbox, like button, 360° tour, gallery manager and cropper, toasts). The panorama viewer is a small dependency-free WebGL module (resources/js/panorama.js). Tailwind CSS 4 is compiled by Vite into one stylesheet per theme.

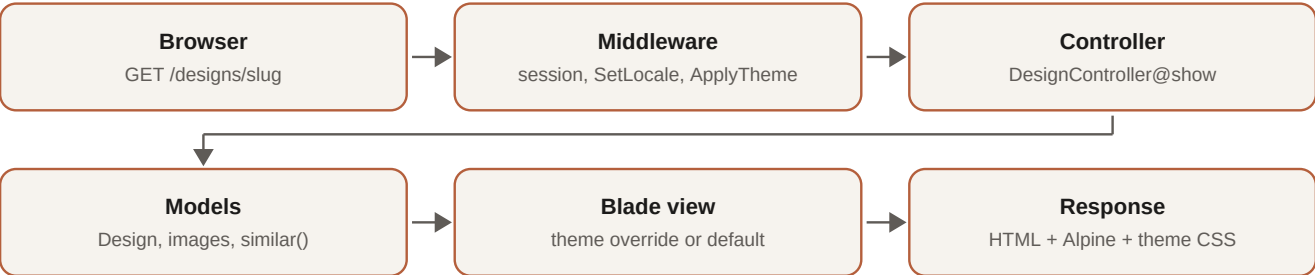


Figure 2. Life of a request.

Two middleware run on every web request:

- **SetLocale** reads the language stored in the session (set by `/lang/{code}`) and applies it, so every `__('ui...')` string renders in English or Simplified Chinese.
- **ApplyTheme** resolves the active theme (a `?theme=name` preview stored in the session beats the admin setting), then puts `resources/views/themes/<name>` in front of the view search path. A theme therefore only needs the views it changes; everything else falls back to the default views.

Key folders

Path	Purpose
<code>app/Models</code>	User, Category (style), RoomType, Design, DesignImage, Order, Setting
<code>app/Http/Controllers</code>	Home, Design (catalogue, detail, suggest, like, download), Auth, Checkout, Dashboard
<code>app/Http/Controllers/Admin</code>	Dashboard, Design, Category, RoomType, Order, User, Theme, Setting, Upload, ImageCrop
<code>app/Http/Middleware</code>	EnsureAdmin, SetLocale, ApplyTheme
<code>app/Console/Commands</code>	<code>images:localize</code> (downloads remote images into the project)
<code>app/Support/helpers.php</code>	<code>price()</code> , <code>thumb()</code> , <code>setting()</code> , <code>available_themes()</code> , <code>active_theme()</code> , <code>compact_number()</code>
<code>app/Support/LocalImage.php</code>	Maps image URLs on this site to files on disk (never fetched over HTTP) and reads their pixel size
<code>resources/views</code>	Default (Calm) theme views, partials and Blade components
<code>resources/views/themes/neon</code>	Neon theme overrides
<code>resources/css</code>	<code>app.css</code> (Calm) and <code>theme-neon.css</code> (Neon), one stylesheet per theme
<code>resources/js/app.js</code>	Alpine store and components shared by both themes, including the 360° tour, gallery manager and cropper
<code>resources/js/panorama.js</code>	WebGL equirectangular panorama viewer: drag, pinch, wheel, keyboard, inertia
<code>lang/{en,zh}/ui.php</code>	Interface strings
<code>public/images/library</code>	All catalogue photos (1600px JPEG + 640px thumbnail)
<code>database/seeder</code>	DatabaseSeeder and SeedData (initial catalogue)
<code>tests/Feature</code>	Download, 360° tour, image crop, theme switch, admin themes and seeder tests

4. Data model

Seven tables carry the whole product. Slugs are generated automatically from names and titles and are what appears in URLs; admin routes also identify designs by slug.

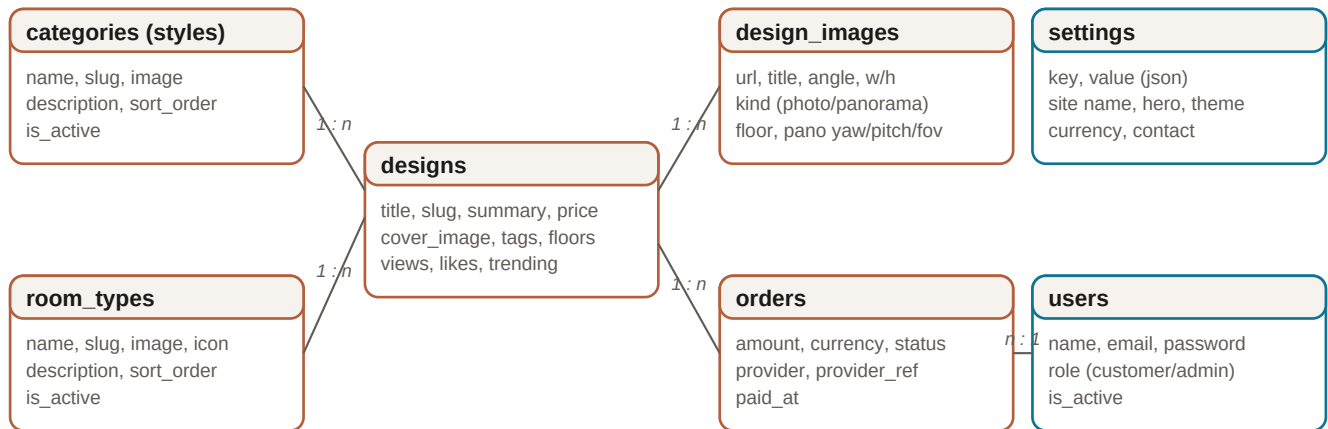


Figure 3. Main entities and relationships.

Table	Important columns	Notes
categories	name, slug, description, image, sort_order, is_active	A design style. Inactive styles disappear from navigation and filters.
room_types	name, slug, description, image, icon, sort_order, is_active	A room. Same rules as styles.
designs	title, slug, summary, description, category_id, room_type_id, tags (json), price, currency, cover_image, designer, area_sqm, featured, published, floors (json), views, purchases, likes, trending_score	price 0 = free. floors holds the floor names used by the 360° tour (default Ground floor). Unpublished designs are hidden from the public site but still editable in admin. trending_score is recomputed on every save.
design_images	design_id, url, width, height, title, description, angle, sort_order, kind, floor, pano_yaw, pano_pitch, pano_fov	One row per gallery image. kind is photo or panorama; a panorama belongs to a floor (1, 2 ...) and opens at the stored yaw, pitch and fov. width and height are measured on save, and panoramas that are not 2:1 are kept out of the tour. Order decides the first (always visible) photo.
orders	user_id, design_id, amount, currency, status, provider, provider_ref, paid_at	status: pending, paid, failed, refunded. provider: demo or stripe. A paid order unlocks the design for that user.
users	name, email, password, role, is_active, preferences	role customer or admin. Inactive users cannot sign in.
settings	key, value (json)	Only keys that differ from the built-in defaults are stored; Setting::allValues() merges them and caches the result.

5. Visitor workflows

5.1 Landing page

The landing page is assembled by HomeController from live data, so it always reflects the catalogue: styles and rooms with their design counts, the eight highest-trending designs and the three highest-trending featured designs.

Section	Content and behaviour
Hero	Tagline, headline, sub-headline, call-to-action and stats come from Site settings. The search box offers live suggestions (designs, styles, rooms) after two keystrokes; Enter runs a full catalogue search.
Browse by style	A carousel of style cards: arrows, position dots, keyboard arrows, swipe, auto-advance every six seconds (paused on hover). Each card links to the catalogue filtered by that style.
Trending now	Eight designs ordered by trending score (see section 10).
Shop by room	One tile per room type, linking to the catalogue filtered by room.
Editor's picks	Three designs flagged Featured in the admin.
How it works, testimonials, newsletter	Static copy from the language files; the newsletter form only shows a thank-you toast (no mailing list is connected).

5.2 Catalogue (/designs)

All filtering lives in the URL, so any filtered view can be bookmarked or shared. Parameters combine freely:

Parameter	Values	Effect
q	free text	Matches title, summary, description, designer and tags (each word OR-ed).
category	style slugs, comma separated	Any of the listed styles.
roomType	room slugs, comma separated	Any of the listed rooms.
price	free or paid	Free designs only, or premium only.
tag	tags, comma separated	Designs carrying every listed tag.
featured	1	Editor's picks only.
sort	trending, newest, priceAsc, priceDesc, popular	Default is trending; popular sorts by views.
page	number	Twelve designs per page.

The Calm theme shows the filters in a sidebar (a slide-in drawer on phones) with checkbox-style rows and popular-tag chips; the Neon theme shows the same filters as pill rows above the grid. Active filters appear as removable chips, and Clear all resets everything except the search text.

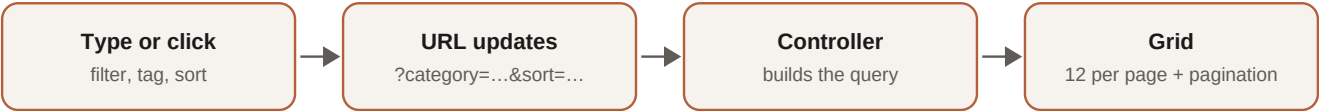


Figure 4. Catalogue filtering.

5.3 Design detail (/designs/{slug})

Opening a design increments its view count and trending score, then renders:

- **Cover and summary:** cover image, style and room links, designer, area, image count, views and the price badge (Free, price, or Unlocked).
- **Purchase card:** price with the free or paid note and the main action. Free or unlocked designs get *Gallery* plus *Download all images*; locked designs get *Unlock full design* (or *Sign in to unlock* for guests).
- **Like and Share:** Like adds one to the public counter and five trending points, turns into *Liked* with a filled heart and is remembered per browser so it cannot be repeated. Share copies the page link and shows a toast.
- **Gallery:** one card per image with its angle badge, title and description. The first image is always visible. Locked images show a blurred cover with a lock and open the checkout when clicked; unlocked images open a lightbox (arrow keys to move, Escape to close) and offer a per-image Download link.
- **360° tour:** one viewer per floor (Floor 1, Floor 2 ...) with a chip for every room panorama. Drag, pinch or scroll to look around; auto-rotate, reset and fullscreen buttons; a viewer only starts when it is scrolled into view. Each room opens at the starting view the admin chose. Floors without a valid panorama are not shown, and on a locked design the tour shows a lock placeholder instead of any panorama.
- **You may also like:** six similar designs (section 10).

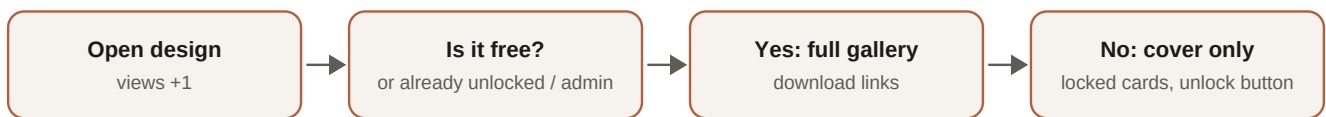


Figure 5. Gallery locking rules.

6. Customer workflows

6.1 Register and sign in

Registration needs a name, email and a password of at least six characters; the account is signed in immediately. Sign in supports *Remember me*. Both forms carry a *next* parameter: when a guest clicks Unlock on a design they are sent to login with next set to that design, and land back on it after signing in. Admins signing in without a next parameter go straight to the admin dashboard. Deactivated accounts are refused with a message.

6.2 Unlock and pay

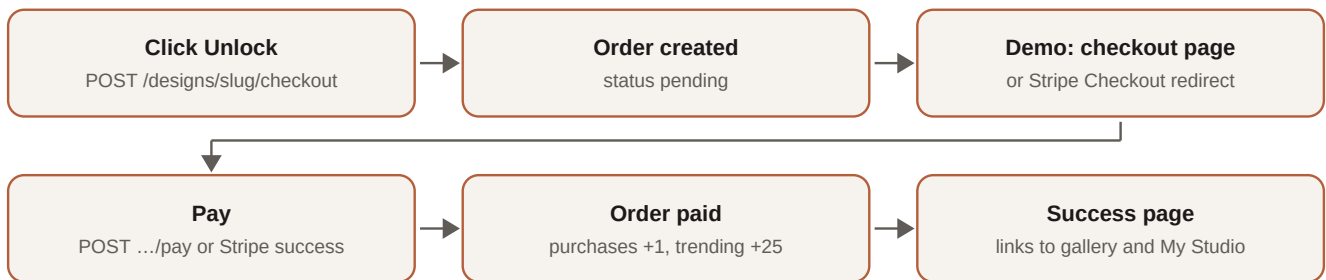


Figure 6. Unlock flow.

- If the customer already owns the design, the checkout redirects back with an *already owned* notice instead of creating a second order.
- A pending order is reused if the customer abandons checkout and comes back, so there is never more than one open order per design and customer.
- The demo checkout page shows the design summary, the total and two buttons: *Pay* (marks the order paid) and *Simulate a declined card* (marks it failed and shows an error toast). No card data is collected.
- With Stripe configured the customer is sent to Stripe Checkout; on return the success page verifies the session with Stripe before marking the order paid. Cancelling returns to the design with a notice.
- The success page confirms the unlock and links to the gallery and My Studio. Reloading a paid order's checkout URL redirects to the success page.

6.3 My Studio (/dashboard)

Tab	Content
Unlocked designs	Cards for every design with a paid order; each opens the full gallery.
Orders	All orders with date, amount and status. Pending orders link back to checkout; paid ones open the design.
Profile	Change display name and password.
Appearance (Calm theme only)	The same theme panel as the header: light/dark/system mode, accent, text and background colours, heading font, text size, corner radius, reset.

6.4 Downloads

For any design the account may fully see, *Download all images* streams a zip named after the design. Photos go in *Photos/* (01-overview.jpg, 02-front-view.jpg ...) and panoramas in one folder per floor (*360 panoramas/Floor 1 - Ground floor/* and so on). Every file keeps its original type, and a README lists the contents. Images on this site are read straight from disk, so the zip is built in about a second. Each gallery card also offers a single-image download. Guests and customers without a purchase receive a 403 on locked designs; this is covered by feature tests.

7. Admin workflows

The admin panel lives at /admin and is available only to accounts with the admin role. It keeps its own look regardless of which customer theme is live. The sidebar lists: Dashboard, Designs, Design styles, Room types, Orders, Customers, Themes and Site settings.

7.1 Dashboard

Stat tiles for revenue (paid orders), designs (published and free counts), customers, orders (with pending or failed count) and total views; a 30-day revenue bar chart with hover tooltips; catalogue mix by room and by style; top designs by sales; and the eight most recent orders.

7.2 Designs

The list is searchable (title, summary, tags, designer) and paginated, with inline toggles for *Published* and *Featured*, links to preview on the site, edit and delete (two clicks to confirm). *Recompute trending* re-saves every design to refresh its score after bulk changes.

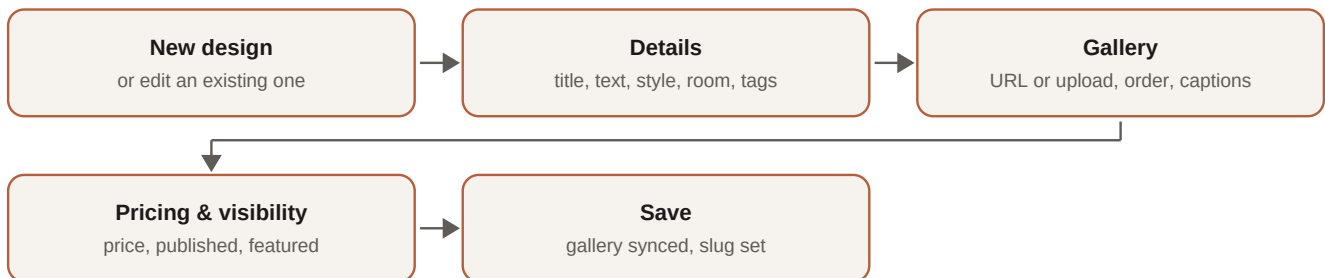


Figure 7. Creating or editing a design.

- **Gallery manager:** paste an image URL and press Enter, or upload one or more files up to 50 MB each (stored under `storage/app/public/designs` and served from `/storage`). Each image has a title, an angle (Overview, Front view, Corner view, Detail, Window side, Night mood, Top view, Elevation, Floor plan) and a description, and can be moved up or down, starred as the cover or removed. On save the gallery is synchronised: kept images are updated, new ones added, removed ones deleted.
- **Photo or 360° panorama:** each image has a Photo / 360° panorama switch. Uploads that are 2:1 and at least 2000 px wide are marked as panoramas automatically. A panorama is assigned to a floor (the Floors card adds, renames and removes floors) and has a *Set starting view* button that opens the viewer so the admin can frame the opening angle. Images marked 360° that are not 2:1 are flagged in the form and on save, and are left out of the tour.
- **Crop:** a crop button on every image, plus an optional crop step right after uploading (the *Crop photos after uploading* checkbox; several uploads open one after another, with Skip this one and Skip the rest). Drag the box to move it and its corners or edges to resize it. Presets are Free, 1:1, 4:3, 3:2, 16:9 and 2:1 · 360°; panoramas are locked to 2:1 so they keep working in the viewer. The browser crops at full resolution and uploads the result as a new file, so the original is never overwritten and the server needs no image extension. Only images hosted on other websites are cropped on the server, which needs PHP gd. The change is kept when the design is saved.
- **Cover:** the starred image or, if none is starred, the first photo.
- **Deleting** a design that has paid orders is refused with a message; unpublish it instead, or delete with force from the URL parameter.

7.3 Design styles and room types

Both pages list the taxonomy with design counts, an inline Active toggle and Edit/Delete actions. A modal creates or edits an entry (name, description, image URL, sort order, and for rooms an icon name). An entry that still has designs cannot be deleted.

7.4 Orders

Filter by status (all, pending, paid, failed, refunded). Changing a status from the dropdown saves immediately; moving an order to paid stamps the paid date and increments the design's purchase count, and moving it away from paid decrements it.

7.5 Customers

Search by name or email; each row shows purchases and total spent. Change a role between customer and admin, toggle Active (inactive accounts cannot sign in) or delete. Admins cannot change or delete their own account here.

7.6 Themes

One card per theme with a colour miniature, swatches, font and mode. *Preview site* opens the customer site in that theme for the admin's browser session only; *Activate for visitors* makes it live for everyone. The live theme carries a Live badge.

7.7 Site settings

Tab	Fields
Branding & landing	Active theme, site name, tagline, announcement bar, footer text, hero headline, sub-headline, button text and image, the three hero statistics.
Theme defaults (Calm)	Default mode (light, dark, system), accent colour, text and background colours for light and dark, heading and body fonts, corner radius, whether visitors may customise their own view. A live preview shows both modes.
Commerce & contact	Currency code and symbol, contact email, Instagram, Pinterest and YouTube links.

Settings are stored as key/value rows and cached; saving clears the cache so the site updates immediately.

8. Payments

Mode	When	Behaviour
Demo checkout	STRIPE_SECRET is empty (default)	An in-app checkout page with Pay and Simulate-declined buttons. No card details, no money. Orders are marked demo.
Stripe Checkout	STRIPE_SECRET is set in .env	The Unlock button creates a Stripe Checkout session (one line item, customer email prefilled) and redirects. The success URL carries the session id; the app retrieves the session and marks the order paid only if Stripe reports it paid. The demo pay endpoint is disabled.

Prices are stored per design with a currency code; the displayed symbol comes from Site settings. Amounts are copied onto the order at checkout time, so later price changes do not alter past orders.

9. Themes

A theme is a folder under resources/views/themes. The default views in resources/views are the Calm theme; the Neon theme (ported from the profile portfolio site) overrides the layout, navigation, footer, landing, catalogue, design detail, dashboard, auth pages, design card, pagination, section header and toasts, and ships its own stylesheet with its own design tokens.

Theme	Look	Mode
Calm (default)	Warm neutrals, serif headlines, visitor-adjustable accent, text and background colours.	Dark by default; light and system available to visitors.
Neon	Near-black surfaces, cyan borders and glow, Inter and JetBrains Mono, pill filters, masthead inner pages, portrait-framed images.	Dark only.

Switching and previewing

- Add **?theme=neon** (or any theme name) to a page URL to preview it for your session; **?theme=site** returns to the live setting. `/theme/{name}` does the same and redirects back.
- The admin Themes page previews and activates themes; the Active theme dropdown in Site settings is the same setting.
- Unknown theme names are ignored (query) or return 404 (route).

Adding a theme

Create `resources/views/themes/<name>/` and copy in only the views you want to change (start with `layouts/app.blade.php`). If the theme needs its own tokens, add `resources/css/theme-<name>.css`, list it in `vite.config.js` and reference it from the theme layout, then run `npm run build`. Optionally describe it in `config/themes.php` so the admin page shows a label, description and swatches. The theme appears in the admin automatically.

10. Trending and similar-design ranking

Trending score

Recomputed whenever a design is saved and nudged live by activity: views (+1 each), purchases (+25 each), likes (+5 each), freshness (+2 per day remaining of the first 30 days) and +40 while Featured. It drives the landing page's Trending section and the default catalogue sort.

You may also like

Signal	Points
Same room type	50
Same design style	40
Each shared tag (up to six)	8
Trending score	up to 20, scaled down to break ties

Every other published design is scored and the top six are shown, so a bedroom in the same style ranks first, then other bedrooms, then other designs of that style. Editing tags or the style of a design in the admin changes its neighbours immediately.

11. Languages and appearance

The interface is available in English and Simplified Chinese (`lang/{en,zh}/ui.php`). The globe menu in the header, or `/lang/{code}`, stores the choice in the session. Catalogue content (titles, descriptions) is entered in one language by the admin and is not translated.

In the Calm theme visitors can personalise their view from the palette icon: mode, accent, text and background colour, heading font, text size and corner radius. Choices are saved per browser and can be reset to the site default. A near-white accent automatically flips to the text colour in light mode (and a near-black one in dark mode) so buttons never disappear. The admin can turn visitor customisation off in Site settings. The Neon theme is fixed dark and ignores these settings.

12. Requirements, setup, seeding, images and tests

Requirements

PHP 8.2 or newer (Composer is pinned to the 8.2 platform, matching production hosting), Composer 2, and Node 20.14 or newer (.nvmrc is 20.14.0; npm refuses older versions because engine-strict is on).

PHP extensions to enable

Enable these in php.ini (on Windows or Laragon, remove the ; in front of the extension= line and restart PHP; on shared hosting, tick them under Select PHP version, Extensions). Most are on by default.

Extension	Needed for	Status
openssl, mbstring, tokenizer, ctype, filter, hash, session, json, pcre	Laravel itself: encryption, sessions, strings	Required
fileinfo	Image uploads: file-type detection and the image validation rule	Required
pdo_sqlite or pdo_mysql	Database (SQLite by default, MySQL optional)	Required
zip	Download all images (zip of photos and panoramas)	Required
curl	Stripe payments and fetching external image URLs	Required
dom, xml, libxml	HTML and CSS inlining for mail, and the test suite	Required
gd	images:localize thumbnails, server-side crop of images on other websites, image fakes in the tests	Recommended
xmlwriter, phar	PHPUnit and dev tools	Dev only

Minimal php.ini lines, plus limits sized for large panoramas:

```
extension=curl
extension=fileinfo
extension=gd
extension=mbstring
extension=openssl
extension=pdo_sqlite ; or extension=pdo_mysql
extension=zip

upload_max_filesize = 50M
post_max_size = 100M
memory_limit = 256M
max_execution_time = 120
```

Check with `php -m` (loaded extensions) and `composer check-platform-reqs --no-dev` (what the installed packages need). Cropping works without gd because the browser does it.

Fresh install

```
composer install
npm install
cp .env.example .env && php artisan key:generate
php artisan migrate:fresh --seed
php artisan storage:link
npm run build
php artisan serve
```

Environment

Variable	Meaning
DB_CONNECTION / DB_DATABASE	SQLite file by default (database/database.sqlite); switch to MySQL with the usual DB_* values.
STRIPE_SECRET	Enables Stripe Checkout when set; empty means demo checkout.
ADMIN_EMAIL / ADMIN_PASSWORD	Credentials the seeder gives the admin account.
APP_URL	Used for absolute URLs, including Stripe return URLs.

Seeding

DatabaseSeeder creates the admin and five demo customers, 10 styles, 10 room types, 45 designs with 3 to 7 gallery images each (207 images), 41 demo orders and one guaranteed unlock for the demo customer. It is idempotent: running it again on a seeded database only re-ensures the accounts. Photos are royalty-free Unsplash images stored in the project. The five CC0 test panoramas from Poly Haven (public/images/library/pano-*.jpg, 4096 x 2048) were attached through the admin to Verdant Haven Lounge (two floors) and Tatami Rest Suite, so a fresh seed does not include them.

Images

All catalogue photos live in public/images/library as 1600px JPEGs with a 640px thumbnail twin used by cards. Admin uploads and crops go to storage/app/public/designs, and images on this site are always read from disk, never fetched over HTTP. If any image is ever referenced by a remote URL, *php artisan images:localize* downloads it into the library, writes the thumbnail and rewrites the database; it is safe to re-run.

Tests

php artisan test runs 32 feature tests: zip downloads, their folder layout and authorisation, the 360° tour (floors, locking, misshapen panoramas), image cropping (browser upload, server crop, clamping, access), theme previews and activation, the admin Themes page, and a seeder guard that seeds a fresh database, checks counts and accounts, verifies every seeded image exists on disk, renders the key pages in both themes and signs in with both demo accounts.

13. Route reference

Method	Path	Purpose
GET	/	Landing page
GET	/designs	Catalogue with filters (section 5.2)
GET	/designs/suggest?q=	JSON search suggestions
GET	/designs/{slug}	Design detail
GET	/designs/{slug}/download	Zip of all images (if unlocked)
POST	/designs/{slug}/like	Like (JSON)
POST	/designs/{slug}/checkout	Start unlock (auth)
GET	/checkout/{order}	Demo checkout page (auth)
POST	/checkout/{order}/pay	Demo payment (auth)
GET	/checkout/success	Success page; verifies Stripe session when present
GET	/dashboard?tab=	My Studio: purchases, orders, profile, appearance
PUT	/dashboard/profile	Update name or password

Method	Path	Purpose
GET/POST	/login, /register	Sign in, create account (with next redirect)
POST	/logout	Sign out
GET	/lang/{code}	Switch interface language
GET	/theme/{name}	Preview a theme for the session (site = back to live)
POST	/newsletter	Newsletter thank-you
GET	/admin	Dashboard (admin)
resource	/admin/designs	List, create, edit, update, delete; PATCH .../toggle; POST .../recompute-trending
resource	/admin/categories, /admin/room-types	Styles and rooms
GET/PATCH	/admin/orders	List and change status
GET/PATCH/DELETE	/admin/users	Customers
GET/POST	/admin/themes, /admin/themes/{name}/activate	Theme preview and activation
GET/PUT	/admin/settings	Site settings
POST	/admin/upload	Image upload for the gallery manager (JSON)
POST	/admin/images/crop	Save a cropped image as a new file (JSON)